2022 IEEE 3rd Global Conference for Advancement in Technology (GCAT)
Bangalore, India. Oct 7-9, 2022

A New Adaptive Modeling and Denoising of Real ECG signal

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***Abstract* - Heart problems are increasing day by day and Electrocardiogram (ECG) signal is extremely important to evaluate cardiac problems. A common problem in ECG interpretation is noise removal because real ECG signals are corrupted by various types of noises and artifacts with different frequencies. Among all the noises the power line interference is the most significant and can strongly affect the ECG signal. This paper presents second order FIR Notch filter to remove power line interference which is occurred with frequency of 50Hz in India, followed by a novel parametric representation of real ECG signals. First, Notch filter is designed here for two cases first one is filter with two zeros, and in second case adding two poles at different radius but in same frequency as those of zeros, but the radius is less then unity in all the cases to consider the stability of filter. The effect of poles with their variables radius (0.8o to 0.99) is also analyzed at the output with SNR and has considered best result in term of stability, magnitude and phase response. Second part includes Signal acquisition, Control point detection, and develops a spline model of real ECG signal.**

Key words: ECG, Notch filter, FIR, power line interference, SNR, Cubic spline, artifact.

I. INTRODUCTION

ECG is a non-stationary biomedical signal that represents the electrical activity of the heart. Analysis of ECG signals gives essential diagnostic information about the heart diseases and any abnormality of heart functioning is reflected in ECG [2, 13]. An exemplary ECG signal consists of repetitively occurring series of beats. The features of ECG signals are depicted by wave peaks P, Q, R, S, T, and U, intervals P-R, R-R, Q-T and segments P-R and S-T [1, 9]. The span between two waves of an ECG is known as a segment, the time duration inclusive of one or more waves and a segment corresponds to an interval as shown in Figure1[2].

Computer Added Diagnostic (CAD) system is based on biomedical signal, this system is the combination of signal acquisition, signal processing and signal analysis. Noise removal is the first building block of signal processing that plays vital role in computer based biomedical systems [10]. ECG is an electrical signal with low amplitude (0.05mv to 5mv) and low frequency range (0.5Hz-100 Hz) generated from high impedance human body. ECG signal stability and randomness depends on the change in human environment and the difference between different individuals [13]. When real ECG is recorded it may be contaminated by various kinds of noises and artifacts that degrade the quality of ECG signals [14].

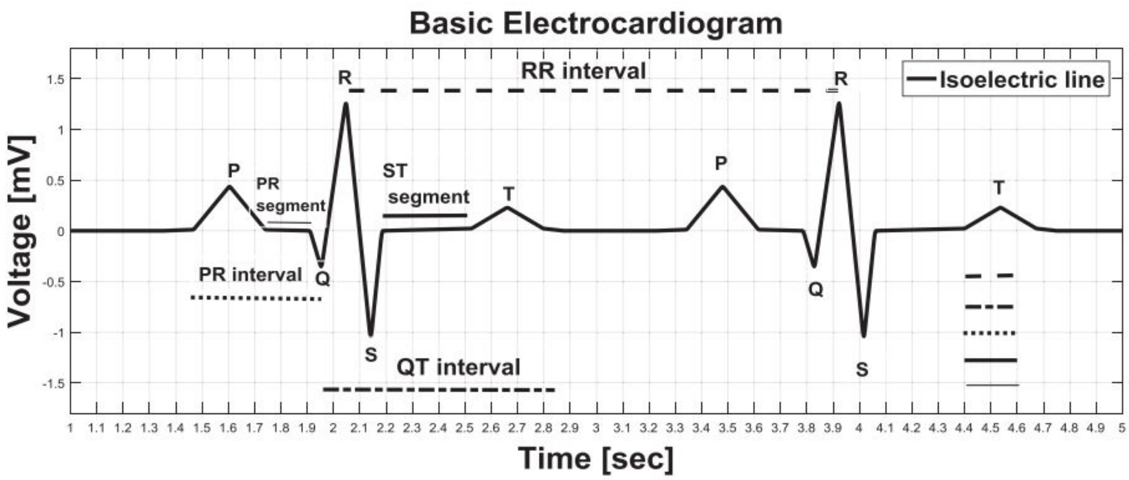

Fig. 1. Normal features of an ECG signal

II. NOISES IN ECG

- Interference due to power line from the neighboring equipment
- Noise due to measurement using electrodes.
- Noise due to electro myographic or muscle contraction.
- Artifacts due to movement
- Respiratory and baseline drift artifacts and
- Noise due to instrumentation.

However, the main sources of interference in ECG signal is power line interference (50-60 Hz) as because it lies ECG signal band consequently it will create the problem to diagnosis arrhythmias [3,7].

The use of signal filtration is one of the options to deal with vitiated signals. Various denoising approaches based on wavelet transform, fuzzy logic, window techniques, FIR filtering, modal based filtering, Adaptive filters etc. has been proposed[4,7].With the advancement of computing technologies, digital filters are extensively applied in signal enhancement, speech recognition and channel image processing [5]. Digital filters are flexible and can be used in very low frequency. Digital filters are classified into two types which are based on to their impulse responses extent i.e. finite impulse response (FIR) and infinite impulse response (IIR). IIR filters are recursive filters, while FIR filters are non-recursive [6].There are various time domain and frequency domain de-noising approaches are used in filters. The time domain filtering has the biggest advantage that it is fast and spectral information about the noise is not required. The only disadvantage is that there is no control on how it will work means cannot actually customize it. Frequency domain filtering is very effective for periodic artifacts, power frequency noise is completely periodic in nature and that kind of artifact can be removed by the frequency domain filter very effectively [7,8]. Notch filter is one of the best methods to reject the signal frequency, such as 50 Hz power line frequency hum. The name of the filter is due to the characteristic shape of its frequency response curve [11].

III. DESIGN AND IMPLEMENTATION OF NOTCH FILTER

This paper propose finite impulse response Notch filter with a sharp phase change and very narrow band property to obtain the noise reduction without introducing the phase distortion. A recorded data (record nsrdb/16265) with sampling frequency 128 Hz is used as a real ECG signal from the benchmark MIT-BIH database. Power line interference of 50 Hz is generated by using MATLAB2016 and noisy ECG signal is obtained by adding the real ECG signal with generated interference signal. To remove the 50 Hz power line interference, the noisy ECG signal is then pass through notch filter. Real and noisy ECG signal is shown in Fig 2 and Fig3 respectively.

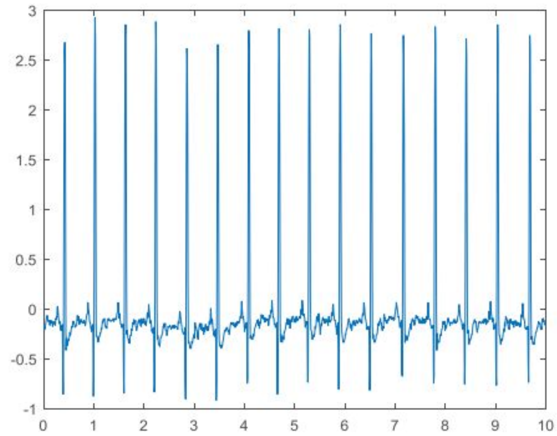

Fig. 2. Real ECG signal

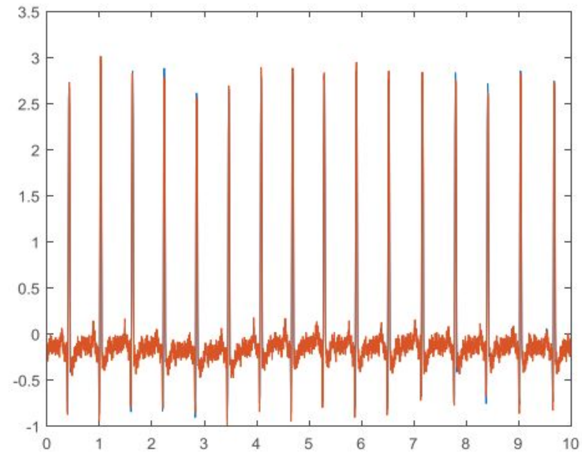

Fig. 3. Noise ECG signal

Notch filter means to set zero at particular frequency (cut off frequency). To design and implement for this filter first set the zeroes on the unit circle at a cut off frequency because zeros will force that the any component present at 50 hertz to zero and instead of one single zero, we are taking two zeros which are actually conjugate pair because our signal is real and we want the output of the filter also should be real. In first case a notch filter designed with two zeros to remove the artifact and implementation is done in MATLAB.

Location of zeros is given by:

$$r\cos w_o \pm j\sin w_o \quad \text{Where; } w_o = \pm 2\pi \frac{f_o}{f_s}$$

Here f_0 is the frequency of noise (power frequency noise) and f_s is the sampling frequency

Zeros in form of Transfer function:

$$H(z) = (1 - z^{-1}z_1)(1 - z^{-1}z_2)$$

Here zeros is assigned at 50 Hz frequency because zeros will force the frequency component to zero which is present at this frequency (50Hz).

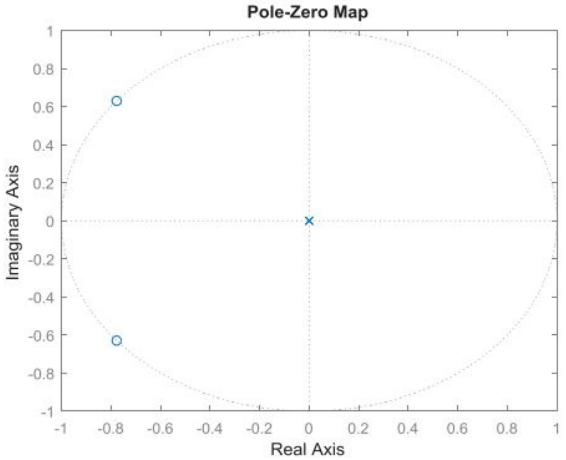

Fig. 4. pole zero plots

Now the pole zero plot shown in Fig4 have the zeros in unit circle as expected and pole at origin zeros are at 141 degree which is more than 90 with respect to zero axis, so they are in the left half of the plane and polynomials here in terms of Z inverse.Fig5 shows the frequency and phase spectrum of notch filter with only two zeros.

If the poles are added at deferent radius from 0.80 to 0.99, as the radius of poles increase the gap between poles and zeros will decrease. At 0.99 radius it Is very difficult to find out the difference between them, it seems they are cancelling each other. But they should not cancel each other if they will cancel the filter become all pass filter. Figure 6 shows how frequency and magnitude plot will change with addition of poles. Here notch become more narrow means adjacent frequency is attenuated.

Poles and zeros in form of Transfer function

$$H(z) = \frac{(1-z^{-1}z_1)(1-z^{-1}z_2)}{(1-z^{-1}p_1)(1-z^{-1}p_2)}$$

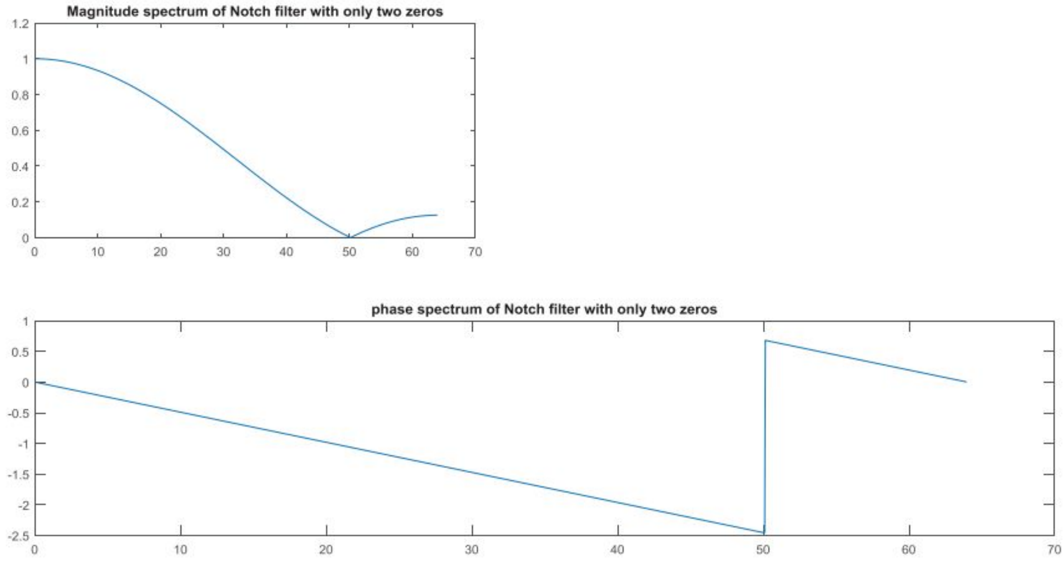

Fig. 5. Frequency and Phase spectrum of filter (with only two zeros).

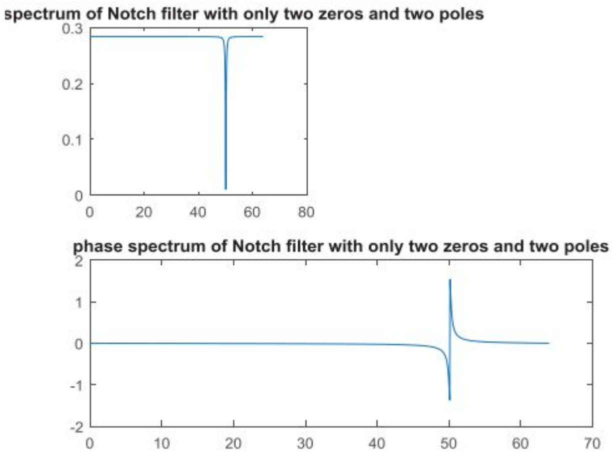

Fig. 6. Frequency and Phase spectrum of filter (with two zeros and two poles)

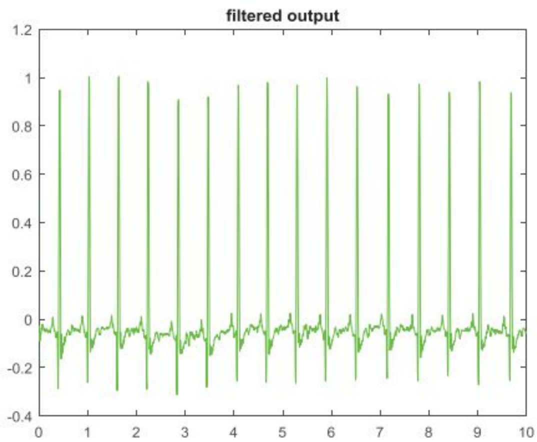

Fig. 7. Filtered ECG signal

The impact of filter can be observed by signal to noise ratio (SNR), which is the ratio of signal energy by noise energy. The Signal energy is energy of filtered output and noise energy is the energy of difference of noisy and filtered signal [12]. Filtered output is shown in fig7. Filter performance parameter (SNR) calculation for different conditions are shown in table1.

$$SNR = 10log_{10}\frac{Signal\ Energy}{Noise\ energy}$$

TABLE I. SNR CALCULATIONS

Sr.No	No of zeros	No of pole	Zeros Radius	poles Radius	SNR
1	2	0	1	No pole	4.2801dB
2	2	2	1	0.8	-5.7219dB
3	2	2	1	0.90	-7.0395dB
4	2	2	1	0.99	-8.1154dB

Many techniques have been proposed to analyse and process ECG signals. It is difficult to check the performance conspicuously because actual ECG is corrupted by several sources of noises and artifacts. No one method is available which can eliminate all the noises and artifacts from the ECG signal. In view of this, generation of artificial ECG signal that can able to reflect all the characteristics of real ECG signal is very challenging task in biomedical signal processing.

IV. PARAMETRIC CUBIC SPLINE CURVE:

Parametric spline curves defined as a piecewise polynomial curves with a certain order of continuity. Parametric cubic splines are used to interpolate to given data points. Actually, it connects two data end points by applying a cubic equation. The parametric equation of a cubic spline segment is given by [15]

$$p(u)= \sum_{i=0}^{3} C_i\, u^i \ ,0 \le u \le 1 (2.1)$$

Where, $P(u)$is position vector of a point on the curve represented by $\vec{p}(u) = [x\ y\ z]^T$and it is a function of u in parametric space and represented by a point vector for Cartesian space. Eq. (2.1) is a function of parameter$\boldsymbol{u}$ and this is used for all, when written for Cartesian space, C_i are the polynomial coefficients. In order to find polynomial coefficient for cubic Hermite curve, following four condition are required first point and last point on the curve i.e. endpoints $(p_0\ , p_1)$ and slope at the points i.e. (p_0', p_1').The tangent vector at any point on the curve can be calculated by differentiating the above equation (2.1) with respect to u and after applying the boundary condition (p_0,p_0'at $u = 0$ and p_1, p_1' at $u = 1$) for cubic spline curve at its endpoints $(p_0\ , p_1)$ the final form of the equation (2.1) for cubic Hermite curve will be

$$P(u) = (2u^3 - 3u^2 + 1)p_0 + (-2u^3 + 3u^2)p_1 + (u^3 - 2u^2 + u)p_0' + (u^3 - u^2)p_1',$$

$$0 \le u \le 1 \qquad (2.2)$$

Where $p_{0,}\ p_1, p_0'\ _, p_1'$geometric coefficients and tangent vector are becomes

$$p^{\cdot}(u) = (6u^2 - 6u)p_0 + (-6u^2 + 6u)p_1 + (3u^2 - 4u + 1)p_0' + (3u^2\ 2u)p_1', \qquad 0 \le u \le 1$$

(2.3)

The above equation (2.2) represents only one cubic spline segment but it can be generalized for any two adjacent segments of a spline curve that are to fit a given number of data points. For given set of n data points, the two end tangent vectors connect the points with cubic spline curve but the tangent vectors at the intermediate points are needed. To eliminate the need of these vectors, the continuity of the curvature at these points can be imposed. To connect two Hermite spline curves to form a second order continuity, the second derivative at the end of the first curve must be equal to the second derivative at the beginning of the first curve. Thus we have:

$$p^{''}(u_1 = 1) = p^{''}(u_2 = 0) \qquad (2.4)$$

Where the subscription of u denotes the segment number, Using this relation, we can further derive the tangent vector at the end of the second curve, which is also the tangent vector at the beginning of the second curve ,by differentiating the Eq (2.3) and using result of Eq (2.4) we get

$$p_1' = \frac{1}{4}(3p_0 + p_0' - 3p_2 + p_2' \qquad (2.5)$$

The unknown tangent vector or slope can be determined by Eq(2.5).For more segments, same procedure will be repeated and a matrix equation is formed. By solving the matrix, intermediate tangent vectors can be determined [15].

In order to do this here areal ECG wave is taken from Physionet data base, and 13 control points have been chosen depending upon the diagnostic importance parametric spline curve has been constructed. Visual form of all 13 control points and modelled signal shown in Fig 8 and Fig 9 respectively.

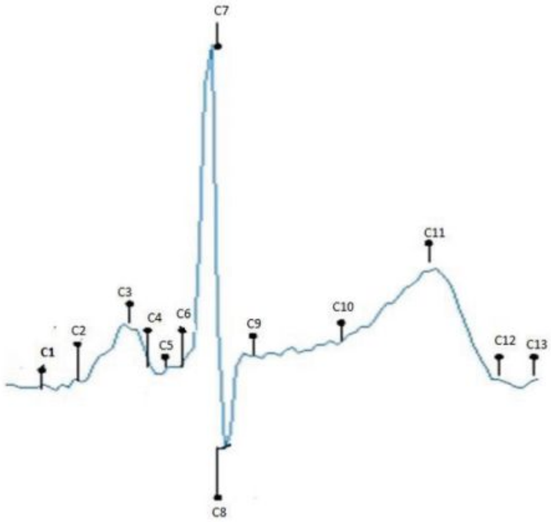

Fig. 8. Visual form of control points

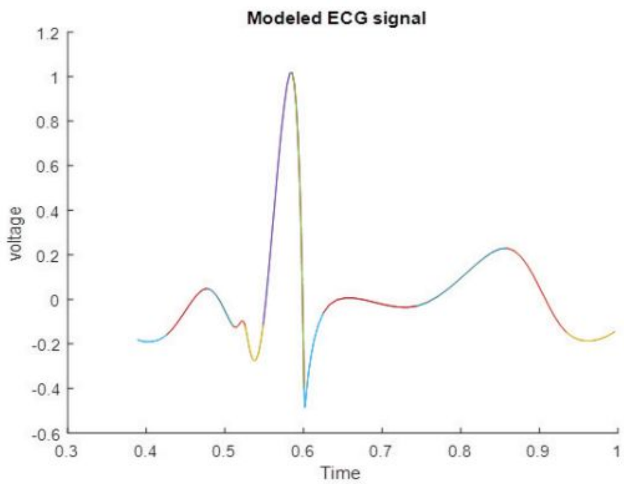

Fig. 9. Modeled ECG signal

For diagnosis of any cardiac disorder, the identification of ECG intervals, segments and location of waves is very necessary.

V. Conclusion

Notch filter with only zeros have wide notch width means this filter will pass some adjacent frequency with noise frequency. This filter performance improved by adding the poles in unit radius. As the location of pole near to zero the notch of the filter become narrower and discontinuity in phase become sharper. It has been observed that notch filter become very close to ideal notch filter at pole radius is 0.99 and phase also has vey sharper change. SNR is also computed to check the efficiency of filter, at 0.99 pole radius highest SNR value achieved. Further research should be done in this area to remove all types of noise and artifacts of ECG signal with single hybrid approach.

In the second part an efficient parametric representation of ECG signals using cubic splines was proposed. The parametric cubic spline provides curvature continuity as well as slope continuity. It is implemented on minimum number of data points, these considered as control points. In some dominions they represent characteristic points, indicating crux changes reflected by the signal. Information regarding exact locations of crucial points provides a deep perspective for diagnosis, analysis and therapeutic purposes.

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